# Forces on two objects

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Identify an equal-and-opposite interaction pair on different objects.

**SUCCESS CRITERIA - I can...**

1. Name both objects in an interaction pair.
2. Place the two equal-and-opposite forces on different objects.
3. Explain why equal forces can produce unequal accelerations.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “interaction” and “system” to explain your model or result. Sentence starters are optional.

**interaction**: two objects affecting one another

**system**: the object or objects chosen for analysis

**acceleration**: rate of change of velocity

## 01 Warm-up: bring back an earlier idea

A 0.4 kg system accelerates at 0.6 m/s².
**Predict** acceleration for 0.8 kg at the same net force.
**Explain** the effect of greater resistance.

**Sentence starter:** Both ... . They differ in ... because ... .

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** $F_{A \text{ on } B} = -F_{B \text{ on } A}$.
The two forces act on different objects at the same instant. Their accelerations can differ.

**Worked reasoning:** Two skaters push apart. Each feels 20 N from the other. If their masses are 40 and 80 kg, accelerations are 0.5 and 0.25 m/s² in opposite directions.

**Your turn: Draw** separate force diagrams for the horizontal interaction.
**Explain** why equal forces do not imply equal accelerations.

**Sentence starter:** Under the assumption ..., the evidence ... supports ... because ... .

Skater A: 40 kg

Skater B: 80 kg

**Marzano check - circle:** ① ② ③

## 03 Equal forces, different results

On a low-friction track, a 1 kg cart and a 4 kg cart push apart. Each interaction force is 12 N.
**Draw** the force on each cart, calculate both acceleration magnitudes, and explain why the forces match but the accelerations do not.

**Sentence starter:** Under the assumption ..., the evidence ... supports ... because ... .



Cart A: 1 kg

Cart B: 4 kg

**Marzano check - circle:** ① ② ③

## 04 Identify actual partners

For a book on a table, name the partner of Earth-on-book and the partner of table-on-book.
**Explain** why weight and normal are not a third-law pair.

**Sentence starter:** Under the assumption ..., the evidence ... supports ... because ... .

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Horse plus cart

A horse pulls a cart forward while the cart pulls backward on the horse with equal force.
**Draw** a boundary around the two-object system. Add the ground's force on the horse and any resistance.
**Explain** how the combined system can accelerate forward.

**Sentence starter:** Under the assumption ..., the evidence ... supports ... because ... .



Horse



Cart

**Marzano check - circle:** 1 2 3

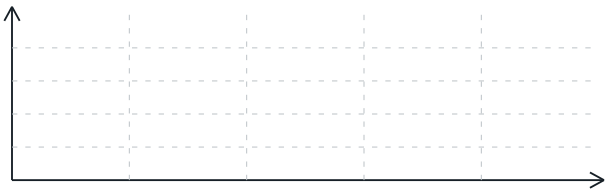
## 06 Review: connect an earlier idea

East is positive. Start at 7 m, go to -5 m, then finish at 4 m.
**Sketch** the route; calculate distance and displacement. Distinguish both from final position.
**Label** the model with the given positions and any prediction.

**Word help:** displacement = change from initial position to final position, with direction

**Sentence starter:** I predict ... . My model supports this because ... .

Route



Position (m)

**Marzano check - circle:** 1 2 3

## 07 Exit check

A spacecraft pushes an asteroid that has much greater mass.
**Draw** the interaction force on each object.
**Compare** their forces and the acceleration caused by this interaction.
**State** what you assume about other forces.

**Sentence starter:** Both ... . They differ in ... because ... .



Spacecraft



Asteroid

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the model

A 2 kg cart pushes a 3 kg cart along a frictionless track. An external 10 N force pushes the 2 kg cart to the right.
**Draw** separate force diagrams, find their shared acceleration, and calculate the contact force.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.



2 kg; applied 10 N right

3 kg

**Marzano check - circle:** 1 2 3